

Tanishq Annavaram

848-313-9205 | tra53@scarletmail.rutgers.edu | www.linkedin.com/in/tanishq-annavaram-86a139227 | <https://github.com/Tanishq111111111>

EDUCATION

Rutgers University - New Brunswick, NJ

Bachelor of Science in Computer Science

Relevant Coursework: Computer Architecture, Intro to **AI**, Database Management Systems(**DBMS**), Data Science, Algorithms

Dec 2026

3.75 GPA

TECHNICAL SKILLS

Languages: Python, R, C/C++, SQL, Java, JavaScript, HTML/CSS, Kotlin, PHP, XML,

Frameworks: PyTorch, TensorFlow, Node.js, Express, React

Databases: MongoDB, MSSQL, MySQL

Libraries: Pandas, NumPy, Matplotlib, Scipy, Scikit-learn, Streamlit

EXPERIENCE

Data & Machine Learning Intern

Algorithmic Control and Robotics Lab

Mar. 2026 – Present

New Brunswick, NJ

- Developed a transformer-based next-state prediction framework for robotic manipulation, extending prior DIPN work on multi-object push interaction modeling.
- Built object-centric learning pipelines using RGB-D inputs, simulation data, and structured object representations to model how pushes propagate across multiple objects in clutter.
- Supporting experimental analysis through attention-map visualization, trajectory analysis, and IoU / manipulation-efficiency evaluation within a broader robotic manipulation pipeline

Research Assistant

Center of Advanced Infrastructure and Transportation(CAIT), Rutgers University

Aug. 2025 – Present

New Brunswick, NJ

- Supported a YOLO-based computer vision pipeline for railroad trespassing detection through video preprocessing, annotation, validation, and QA on multi-site real-world footage.
- Performed error analysis on model outputs and converting raw detections into structured datasets and short evidence clips for downstream analysis
- Developed clear technical documentation and data guides for research teams and non-tech staff, improving workflow consistency and reducing clarification requests by 30%.

Software Engineer Intern

BizBites Technologies

Apr. 2025 – Aug. 2025

Remote

- Built a secure backend with Node.js, Express.js, and MySQL, implementing bcrypt-based authentication that reduced login errors by 40%.
- Developed RESTful APIs for device/switch management with transactional queries, improving database performance by 35% and cutting device response latency by 25%.
- Streamlined deployment using Docker, Postman, and GitHub Actions (CI/CD), accelerating release cycles by 50%.

PROJECTS

LLM-Factcheck | *Python, SQL, Pandas, NumPy, Scikit-learn, Hugging Face Datasets, BM25, FAISS, OpenAI API, Streamlit*

- Built a reproducible Python evaluation framework for factual QA across TriviaQA and Natural Questions, comparing 3 systems: direct LLM prompting, retrieval-grounded generation, and extractive search baselines.
- Designed reliability metrics using Pandas, NumPy, Scikit-learn, and sentence-transformers, measuring exact match, token F1, semantic similarity, groundedness, hallucination rate, and confidence-correctness alignment.
- Created dashboard-ready experiment outputs and error analysis reports, showing retrieval grounding improved factual QA reliability by 40% and identifying 400 key failure modes through structured taxonomy and manual review.

Splitshare-Expense Splitting Tool | *JavaScript, Node.js, Express.js, React.js, MongoDB, REST API design*

- Engineered a full-stack expense-sharing application (MERN stack) that automated group bill settlements, reducing manual calculation time by 80%.
- Optimized debt resolution using a greedy $O(n \log n)$ algorithm, decreasing required transactions by 50–70% in test groups of 10+ users.
- Built and documented RESTful APIs in Express.js with MongoDB, improving response efficiency by 40%.